

Mathematically Exact Formulation of Spectral Flux and Adaptive Peak Picking for Vocal Percussion Segmentation in C++ WebAssembly Environments

The Acoustic Phenomenology of Vocal Percussion and Time-Domain Limitations

The automated segmentation of human vocal percussion, colloquially known as beatboxing, presents a digital signal processing (DSP) challenge that vastly exceeds the complexity of segmenting traditional acoustic or synthetic percussion. Traditional percussive instruments produce sound via the collision of rigid or semi-rigid bodies—such as a drumstick striking a synthetic membrane or a beater striking a brass cymbal. These physical interactions generate nearly instantaneous acoustic transients characterized by a sudden, massive injection of broadband energy. Consequently, the time-domain envelope of a traditional drum hit features an abrupt vertical spike, making it highly amenable to segmentation via basic volume thresholding or elementary envelope followers.

Conversely, human vocal percussion relies upon the highly deformable, non-rigid tissues of the vocal tract, lips, tongue, and glottis.¹ The acoustic signatures of beatboxing can be divided into several phonetic categories, each exhibiting wildly divergent temporal and spectral behaviors. Plosives, which form the basis of vocal kick drums and basses, are generated by occluding the vocal tract, building subglottal air pressure, and releasing it through the yielding tissue of the lips. Because the lips are pliable, this release is not instantaneous. The resulting acoustic waveform presents a low-frequency oscillation coupled with a moderately smeared broadband click, often characterized by a gradual attack slope lasting between ten and twenty milliseconds.¹

Fricatives and affricates, which form the basis of vocal hi-hats, cymbals, and shakers, are produced by forcing a continuous stream of air through a narrow constriction, such as between the tongue and the alveolar ridge. This mechanism generates high-frequency, noisy turbulence.² Acoustically, vocal fricatives possess an extremely low overall root-mean-square (RMS) amplitude when compared to voiced sounds or plosives, yet they contain intense spectral energy strictly localized in the upper-frequency bands (typically above 5 kHz). Complex compound sounds, such as vocal snare drums, merge a sharp plosive release with a prolonged fricative noise tail, creating dual-stage transients that frequently cause naive detection algorithms to register false positive double-hits.

Due to this extreme acoustic variance, standard time-domain thresholding fundamentally fails. A loud, sustained vocalization, such as a throat-bass, will continuously exceed a static volume threshold, while a breathy, unvoiced hi-hat will fail to register entirely.³ The requisite solution, strongly established in Music Information Retrieval (MIR) literature, demands a transition into the frequency domain.⁵ By utilizing a Spectral Flux algorithm paired with Dynamic Adaptive Peak Picking, a DSP engine can measure the relative rate of change of energy across individual frequency bins rather than relying on absolute broadband amplitude.⁶ This paradigm shift ensures that a quiet vocal hi-hat is reliably detected because, relative to the preceding silence in the high-frequency bands, the injection of energy represents a mathematically massive rate of change.

This document serves as an exhaustive, mathematically rigorous blueprint for implementing a dependency-free, C++ based Spectral Flux onset detection engine explicitly optimized for compilation to WebAssembly (WASM). The analysis is strictly confined to the continuous algorithmic processing of floating-point arrays, defining the precise mathematical logic required to achieve highly deterministic hit-point segmentation without reliance on deep learning architectures or external computational libraries.

C++ WebAssembly Architectural and Memory Constraints

Before detailing the mathematical transformations, the architectural constraints of the target environment must be established. WebAssembly modules operating within the browser execute in a sandboxed environment where memory is represented as a contiguous, resizable `ArrayBuffer`. In high-performance, real-time or near-real-time DSP applications, dynamic memory allocation during the audio processing loop (e.g., utilizing `std::vector::push_back` or allocating heap memory via `new`) introduces unacceptable latency spikes and non-deterministic execution times due to potential garbage collection overhead or memory reallocation.

Therefore, the C++ implementation must rely on a strictly pre-allocated memory architecture. All required state arrays—including the circular buffers for incoming audio, the windowing function coefficients, the previous frame's magnitude spectrum, and the scratch buffers necessary for the median filtering algorithms—must be allocated during the engine's initialization phase. The data flow consists of raw floating-point audio samples arriving from the Web Audio API into a pre-allocated input buffer. The engine processes this data in overlapping chunks, advancing a read pointer, and ultimately populating a pre-allocated output array of 32-bit unsigned integers representing the exact sample indices of the detected hit points.

The algorithms detailed in the subsequent sections are designed specifically to operate within this zero-allocation paradigm, ensuring that the computational complexity remains rigidly bounded and that the memory footprint remains static throughout the lifecycle of the analysis.

Signal Framing and the Time-Frequency Uncertainty Principle

To transition the continuous time-domain audio signal into the frequency domain, the engine must employ the Short-Time Fourier Transform (STFT). The STFT operates by dividing the signal into short, overlapping frames and applying a Discrete Fourier Transform (DFT) to each frame. The fundamental mathematical limitation of this approach is governed by the Gabor limit, a manifestation of the time-frequency uncertainty principle.⁸

The Mathematics of Window Size Selection

The precision of the analysis is constrained by a strict tradeoff. An extensive time window containing a large number of samples provides exceptional frequency resolution, allowing the algorithm to finely distinguish between adjacent frequencies. However, a large window heavily smears the temporal location of the transient event, masking the exact moment the onset occurred. Conversely, a narrow time window provides exceptional temporal resolution but severely degrades frequency resolution, rendering it difficult to distinguish a low-frequency vocal kick from generic broadband noise.⁹

For human vocal percussion, the optimal mathematical compromise must resolve the lowest fundamental frequencies of a vocal kick—which can drop to approximately 50 Hz—while maintaining a temporal resolution tight enough to separate rapid rhythmic subdivisions, such as 32nd-note hi-hat patterns that may occur mere tens of milliseconds apart.

Assuming a standard sample rate of $F_s = 44100$ Hz, the frequency bin resolution Δf and the temporal duration T_w of an FFT window of size N are calculated as:

$$\Delta f = \frac{F_s}{N}$$

$$T_w = \frac{N}{F_s}$$

Evaluating the standard powers of two:

- An $N = 512$ window yields $T_w \approx 11.6$ ms and $\Delta f \approx 86.13$ Hz. This temporal resolution is excellent, but the frequency resolution is too coarse to accurately capture the sub-harmonics of a vocal kick.¹⁰
- An $N = 2048$ window yields $T_w \approx 46.4$ ms and $\Delta f \approx 21.53$ Hz. The frequency

resolution is exceptional, but the temporal smearing of nearly 50 milliseconds will destroy the precise timing quantization required for MIDI conversion.¹⁰

- An $N = 1024$ window yields $T_w \approx 23.2$ ms and $\Delta f \approx 43.06$ Hz.

Based on established MIR literature targeting percussive onset detection, $N = 1024$ is **mathematically optimal**.¹¹ A 1024-sample window provides sufficient frequency resolution to capture the critical 40–80 Hz energy burst of a plosive, while its 23.2-millisecond duration restricts the temporal smearing of the onset envelope, facilitating sharper peak detection of fricative hi-hats.

Hop Size and Overlap Implementation

The hop size, denoted as H , dictates the precise number of discrete samples the algorithm advances the analysis window for each successive STFT frame.¹³ The hop size directly establishes the temporal resolution of the final onset detection envelope.

If the hop size is excessively large, the exact sample index of the hit point will be quantized to a coarse grid, fundamentally compromising the rhythmic accuracy of the downstream MIDI conversion. If the hop size is overly small, the computational load increases dramatically, potentially violating the strict execution time constraints of a WASM environment operating within a single browser thread.

A widely adopted standard in MIR research for high-resolution onset detection is a hop size of $H = 256$ samples.¹⁵ At 44.1 kHz, this corresponds to an advancement of approximately 5.8 milliseconds per frame, resulting in a 75% overlap with an $N = 1024$ window. This overlap ensures that no transient energy is lost to the attenuation introduced by the windowing function at the boundaries of the frame.

Windowing Function Mathematics

Prior to executing the Fast Fourier Transform (FFT), the N -sample array must be multiplied element-wise by a windowing function. Failure to apply a windowing function results in spectral leakage, wherein discontinuities at the edges of the finite frame cause energy to bleed across the entire frequency spectrum, obscuring the precise spectral changes the algorithm seeks to measure.¹⁶

While certain specialized applications might utilize Discrete Prolate Spheroidal Sequences (DPSS) or Kaiser windows to maximize the energy concentration of the main lobe¹⁷, the Hann window remains the mathematical standard for onset detection due to its excellent side-lobe attenuation and perfect reconstruction properties when utilizing a 75% overlap.¹⁵

The Hann window function $w(n)$, for sample index $n = 0, 1, \dots, N - 1$ is mathematically defined as:

$$w(n) = 0.5 \left(1 - \cos \left(\frac{2\pi n}{N - 1} \right) \right)$$

Within the C++ architecture, this mathematical function must be pre-calculated precisely once during engine initialization. The resulting coefficients are stored in a contiguous, flat `std::vector<float>`. During the real-time processing loop, the raw time-domain samples copied from the circular input buffer are multiplied element-wise by this pre-computed array immediately before being passed to the dependency-free FFT algorithm.

Windowing Parameter	Mathematical Symbol	Value	Justification
FFT Window Size	N	1024 samples	Balances sub-bass frequency resolution with temporal sharpness. ¹⁰
Hop Size	H	256 samples	Provides 5.8ms temporal resolution and 75% overlap. ¹⁵
Window Type	$w(n)$	Hann	Standardized side-lobe attenuation with efficient overlap characteristics. ¹⁸

The Mathematical Formulation of the Spectral Flux Engine

The Spectral Flux algorithm quantifies the temporal evolution of the audio signal's spectral profile by computing the difference between the magnitude spectra of successive STFT frames.¹⁴ When a vocal percussionist articulates a sound, energy is rapidly injected into specific frequency bands. By isolating and aggregating these positive energy increases, the engine generates a highly responsive "Onset Strength Envelope" (also referred to as a Novelty

Function).⁵

Magnitude Spectrum Extraction and Logarithmic Compression

For an input signal $x(n)$, the complex STFT output at frame m and frequency bin k is denoted as $X_{complex}(m, k)$. The linear magnitude spectrum $|X(m, k)|$ is derived by calculating the Euclidean norm of the real and imaginary components:

$$|X(m, k)| = \sqrt{\text{Re}(X_{complex}(m, k))^2 + \text{Im}(X_{complex}(m, k))^2}$$

A critical nuance in MIR analysis of vocal percussion is the extreme dynamic range between different vocalizations. A vocal kick drum moves a substantial volume of air, resulting in a high-amplitude signal that dominates the linear magnitude spectrum. A vocal hi-hat, composed mostly of forced breath, results in a comparatively low-amplitude signal. If the algorithm processes the linear magnitude directly, the subsequent flux calculation will be overwhelmingly dominated by the low-frequency plosives, and the high-frequency fricatives will register as negligible, undetectable ripples.¹⁴

To rectify this acoustic imbalance, logarithmic compression is applied to the magnitude spectrum.⁵ Logarithmic compression mathematically amplifies low-amplitude high-frequency noise while simultaneously compressing high-amplitude low-frequency thuds, placing both percussive elements on an equitable analytical scale.

The logarithmically compressed magnitude $Y(m, k)$ is calculated as:

$$Y(m, k) = \ln(1 + \gamma \cdot |X(m, k)|)$$

The γ (gamma) variable functions as a compression or scaling parameter. A widely adopted value in MIR research for spectral flux is $\gamma = 10.0$ or $\gamma = 100.0$.⁶ The addition of the constant 1 inside the logarithmic function is a strict mathematical necessity; it ensures that the argument to the logarithm is never less than 1 . If a frame contains absolute digital silence ($|X(m, k)| = 0$), the formula evaluates to $\ln(1) = 0$, gracefully handling silence without generating undefined or infinite floating-point values.

Half-Wave Rectification (HWR) Logic

The objective of an onset detection algorithm is exclusively to identify the precise moment a

musical event *begins*. Therefore, the mathematical logic is solely interested in instances where the energy within a frequency bin has *increased* relative to the previous frame. If the energy decreases—such as during the decay tail of a vocal kick drum or the ambient fading of a room's natural reverberation—this information must be systematically discarded to prevent the algorithm from generating false positive detections during the offset of a sound.⁵

This isolation of positive energy is achieved utilizing Half-Wave Rectification (HWR). In modern computational mathematics, this operation is identical to the Rectified Linear Unit (ReLU)

function commonly employed in artificial neural networks.¹³ For a given differential value x , the HWR function $H(x)$ is defined as:

$$H(x) = \frac{x + |x|}{2} = \max(0, x)$$

In the C++ implementation, this is executed as a simple conditional check or via `std::max(0.0f, difference)` during the evaluation of the spectral bins, ensuring that any negative flux difference is clamped strictly to zero.

The Norm Comparison: L1 vs. L2 in Vocal Percussion

The total Spectral Flux for a specific frame m is determined by aggregating the rectified differences across all usable frequency bins k . The academic literature generally presents two distinct methodologies for this mathematical aggregation: the L1 norm (Manhattan distance, representing the sum of absolute differences) and the L2 norm (Euclidean distance, representing the sum of squared differences).¹⁹

The mathematical formulation for the L2 norm aggregation is:

$$SF_{L2}(m) = \sqrt{\sum_{k=0}^{N/2} (H(Y(m, k) - Y(m - 1, k)))^2}$$

The mathematical formulation for the L1 norm aggregation is:

$$SF_{L1}(m) = \sum_{k=0}^{N/2} H(Y(m, k) - Y(m - 1, k))$$

For the specific application of human vocal percussion, the **L1 norm is mathematically and acoustically superior**.⁶ The L2 norm introduces an exponential weighting applied to large differences within individual frequency bins. If a transient contains a highly resonant, narrow-band whistle or a localized harmonic spike, the L2 norm will square that massive isolated difference, causing it to disproportionately dominate the overall flux signal.

Conversely, the L1 norm treats the absolute difference in every frequency bin with equal linear weighting. A vocal hi-hat (fricative) produces an incredibly small increase in energy per bin, but it does so simultaneously across hundreds of high-frequency bins. The L1 norm simply aggregates these hundreds of minute differences, resulting in a strong, robust onset spike. Furthermore, the L1 norm naturally enforces sparsity within the resultant vector space, which has the practical effect of heavily suppressing scattered background noise and lowering the overall noise floor of the generated Novelty Function.²⁴

Advanced Trajectory Tracking: Vibrato Suppression via SuperFlux

A unique challenge presented by human vocal percussion is the inherent presence of vocal tremolo, vibrato, and pitch instability.²⁹ When a vocalist sustains a note or performs a lip-roll, the fundamental frequency rapidly oscillates. In the frequency domain, this causes energy to continuously shift back and forth between adjacent frequency bins. A standard Spectral Flux algorithm will detect the energy entering a bin as a positive increase, undergo half-wave rectification, and interpret the vibrato as a rapid succession of false positive onsets.²⁹

To mathematically suppress this phenomenon without relying on complex machine learning architectures, the DSP engine can implement a modification of the Spectral Flux algorithm known as **SuperFlux**.¹² SuperFlux replaces the standard frame-to-frame difference with a difference calculated against a trajectory-tracking maximum filter.

Instead of comparing the current frame $Y(m, k)$ directly to the immediately preceding frame $Y(m - 1, k)$, SuperFlux compares it to the maximum value observed in that specific frequency bin over a short preceding history of length L .²⁹ The maximum filter $M(m, k)$ is defined as:

$$M(m, k) = \max_{\mu \in [m-L, m-1]} Y(\mu, k)$$

The SuperFlux equation then becomes:

$$SF_{super}(m) = \sum_{k=0}^{N/2} \max \left(0, Y(m, k) - M(m, k) \right)$$

If the energy in a bin is fluctuating due to vibrato, the maximum filter sustains the peak value across the fluctuation, meaning the difference $Y(m, k) - M(m, k)$ will remain zero or negative, entirely suppressing the false positive. While implementing SuperFlux requires maintaining a circular history buffer of the magnitude spectra for L frames (typically $L = 2$ or 3), it provides unparalleled robustness against vocal instability.¹²

For the baseline "Beatbox2MIDI" prototype, the standard L1 Spectral Flux is sufficient if the performer utilizes short, staccato articulations. However, if the vocalists frequently utilize sustained sounds, implementing the SuperFlux maximum filter is highly recommended. The locked-down mathematical formulation for the standard L1 Spectral Flux to be implemented in C++ is:

$$SF(m) = \sum_{k=0}^{N/2} \max \left(0, \ln \left(1 + \gamma |X(m, k)| \right) - \ln \left(1 + \gamma |X(m-1, k)| \right) \right)$$

Dynamic Adaptive Peak Picking and Thresholding Logic

Once the Spectral Flux array (the Onset Strength Envelope) is fully populated for the audio segment, the system faces the complex task of identifying the exact frame indices where a beatbox hit occurred.

A naive engineering approach would be to apply a static, global threshold (e.g., if

$SF(m) > 0.5$, an onset is flagged). However, human beatbox performances are highly dynamic; a performer may initiate a rhythm by whispering and gradually crescendo into a loud, aggressive climax.³ A static threshold inevitably fails: if set too high, it misses all the whispered hits; if set too low, it registers false positives on ambient breath noise during the loud climax.³³

Therefore, the threshold must be **dynamic and adaptive**, continuously adjusting its sensitivity based on the localized energy of the performance. This local threshold is calculated by analyzing a moving neighborhood (window) of frames symmetrically surrounding the current frame m .

The Mathematics of the Moving Window

The local threshold $T(m)$ at frame index m is derived from a subset of the Spectral Flux array spanning from $m - \frac{W}{2}$ to $m + \frac{W}{2}$, where W represents the adaptive threshold window size in frames.⁵ Because the "Beatbox2MIDI" application processes pre-recorded raw audio chunks (rather than operating as a live, zero-latency real-time effect), the peak picking algorithm can operate non-causally. This permits the algorithm to "look ahead" into future frames to establish the threshold, rather than being strictly limited to past frames.

The literature identifies two primary statistical methodologies used to compute the baseline of this local window: the Moving Mean and the Moving Median.³

Moving Mean Implementation

The Moving Mean calculates the simple arithmetic average of the local window:

$$LocalBase_{mean}(m) = \frac{1}{W} \sum_{j=-W/2}^{W/2} SF(m + j)$$

From a C++ performance perspective, the Moving Mean is computationally trivial. It can be optimized to execute in $O(1)$ time per frame by maintaining a running sum (adding the incoming leading value of the window and subtracting the outgoing trailing value). However, the mean is highly susceptible to outliers. Because an onset peak is, by mathematical definition, a massive outlier relative to the surrounding noise floor, the presence of the peak itself violently skews the mean upward. This "self-masking" effect frequently causes the threshold to rise so drastically that it swallows smaller, adjacent peaks.³

Moving Median Implementation

The Moving Median finds the exact middle value of the sorted local window:

$$LocalBase_{median}(m) = \text{median}\left(SF(m - W/2), \dots, SF(m + W/2)\right)$$

The median is exceptionally robust against outliers.³ The presence of a massive onset peak within the window does not skew the median of the surrounding noise floor. This provides a vastly superior, highly stable baseline that accurately traces the true ambient noise and breathiness of the beatbox recording without being polluted by the hits themselves.

For a WebAssembly application where segmentation accuracy is paramount and the window

size W is relatively small (typically between 10 and 30 frames), the **Moving Median is strongly mandated**. While computing a median mathematically requires sorting data, it can be implemented efficiently in C++ without triggering a full $O(W \log W)$ sort. By utilizing the `std::nth_element` algorithm on a pre-allocated scratch buffer, the median can be extracted in $O(W)$ average time complexity.

The Sensitivity Multiplier and Noise Floor Offset

The computed $LocalBase(m)$ essentially maps the localized noise floor of the Spectral Flux. To qualify as an actual onset, the flux at frame m must exceed this noise floor by a statistically significant margin. This margin is mathematically governed by sensitivity parameters.

The literature presents two standard mechanisms for applying this sensitivity:

1. **Additive Offset (δ):** The threshold is shifted upward by a static constant:

$$T(m) = LocalBase(m) + \delta_{37}$$

2. **Multiplicative Scaling (α):** The threshold is multiplied by a scaling factor:

$$T(m) = LocalBase(m) \cdot \alpha_{39}$$

Because the Spectral Flux algorithm already applies logarithmic compression to the magnitude spectrum, an additive offset in the logarithmic domain is mathematically equivalent to a multiplicative scaling in the linear amplitude domain. While an additive offset is common in simple MIR implementations³, it fails to account for extreme variance. The most robust mathematical formulation for C++ implementation combines *both* an α multiplier and a δ offset³³:

$$T(m) = (\alpha \cdot \text{median}_{j \in SF(m+j)}) + \delta$$

- **α (Sensitivity Multiplier):** This parameter scales the local median.³⁶ A value of $\alpha = 1.5$ requires the onset peak to be 50% larger than the surrounding median noise floor. This multiplier ensures that as the overall intensity of the performance increases, the threshold scales proportionately.
- **δ (Noise Floor Offset):** This is a small, static mathematical constant (e.g., 0.02). Its sole purpose is to prevent false triggers during total digital silence. If the performer stops making sound, the local median drops to exactly 0.0. Multiplying 0.0 by α yields 0.0.

Without the δ offset, the threshold would be zero, and any microscopic floating-point noise or room tone would erroneously trigger an onset.

Threshold Strategy	Mathematical Formula	Robustness to Peak Outliers	Computational Complexity
Moving Mean	$\frac{1}{W} \sum SF(m + j)$	Poor (Self-masking)	$O(1)$ via running sum
Moving Median	$\text{median}(SF(m + j))$	Excellent	$O(W)$ via <code>std::nth_element</code>

Mapping the Threshold to Sample Index i

A critical requirement of the "Beatbox2MIDI" engine is that it must output an array of exact time-domain sample indices corresponding to the precise location of the hits. However, the Spectral Flux and the Dynamic Threshold operate entirely within the downsampled frame domain.

If the algorithm detects an onset at frame index m , it must mathematically map that frame back to the time-domain sample index i .⁶ The frame index m represents a window of N samples that was analyzed. The transient event could have occurred anywhere within that N -sample window.

Given that the Spectral Flux essentially calculates the difference between frame m and frame $m - 1$, the onset energy has definitively entered the signal within the samples introduced by the current hop. However, because the Hann window heavily attenuates the edges of the frame, the energy of the signal is most accurately localized at the absolute center of the FFT window.

The precise mathematical transformation from frame index m to sample index i is calculated as:

$$i = m \cdot H + \frac{N}{2}$$

Where H is the hop size and N is the FFT window size. This equation places the reported hit point exactly at the geometric center of the window that generated the maximum spectral flux peak. This calculation must be applied precisely at the moment a local peak is validated against the threshold.

Refractory Periods and False Positive Lockout Logic

Beatboxing generates acoustic events that are highly irregular. A vocal snare drum frequently features a sharp plosive lip-smack followed immediately by an exhaled burst of air simulating the snare wires. Within the Spectral Flux envelope, this single musical intent manifests not as a single clean spike, but as a jagged, double-peaked mountain.⁴¹

If the DSP engine blindly selects every individual point where the flux $SF(m)$ exceeds the local threshold $T(m)$, a single vocal snare will register as two or three separate MIDI notes fired within ten milliseconds of each other. To prevent this catastrophic quantization failure, the C++ engine must implement a strict **Refractory Period**, frequently referred to in DSP literature as a Minimum Peak Distance.⁴²

Mathematical Logic of Peak Qualification

Before the refractory period is even considered, a specific frame m must first be mathematically qualified as a valid local peak candidate. A frame m is considered a candidate if and only if it satisfies two simultaneous mathematical conditions:

Condition 1 (Thresholding Exceeded):

The flux at frame m must be strictly greater than the dynamically calculated threshold at frame m .

$$SF(m) > T(m)$$

Condition 2 (Local Maximum Identification): The flux at frame m must be strictly greater than the flux at the immediately preceding frame and the immediately following frame.⁵ This ensures the algorithm only triggers on the exact apex of the curve, rather than triggering on

the rising slope of the onset.

$$SF(m) > SF(m - 1) \quad \text{AND} \quad SF(m) > SF(m + 1)$$

Enforcing the Lockout Period at the Sample Level

The refractory period introduces a time-domain lockout constraint to the processing loop.⁴⁵

Once an onset candidate is confirmed and its sample index i is recorded, the algorithm intentionally becomes "blind" to any further onset candidates for a specified duration of time, denoted as $T_{refractor}$ (typically measured in milliseconds).

To execute this efficiently in C++ without relying on time-based chronometers, the temporal value $T_{refractor}$ must be converted into an exact number of discrete array indices (frames).⁴⁴

Given a refractory duration in seconds T_{sec} , a sample rate F_s , and a hop size H , the refractory period in frames ($M_{refractory}$) is mathematically defined as the ceiling of the division:

$$M_{refractory} = \left\lceil \frac{T_{sec} \cdot F_s}{H} \right\rceil$$

C++ State Machine Implementation

The final peak-picking and refractory logic is most efficiently implemented as a single-pass state machine moving sequentially through the populated onset_envelope array. This process strictly manipulates array indices to manage state.

1. **State Initialization:** Define an integer variable `last_onset_frame = -M_refractory`. Initializing this tracking variable to a negative value ensures that a legitimate hit occurring at the very beginning of the audio file is not inadvertently locked out by the initial state.
2. **Iteration:** Loop variable m from 1 to $M - 2$ (where M is the total number of frames, avoiding out-of-bounds access on the $m + 1$ check).
3. **Qualification Check:** Evaluate Condition 1 and Condition 2.
4. **Refractory Evaluation:** If the qualification check passes, evaluate the distance from the last registered onset:
 - If $(m - \text{last_onset_frame}) \geq M_refractory$
 - **Valid Hit Confirmed!**
 - Update the state tracker: `last_onset_frame = m`

- Execute the mapping equation: $\text{hit_sample} = m * H + N/2$
- Append `hit_sample` to the pre-allocated output array.

This exact mathematical sequence guarantees that noisy, jagged peaks generated by breathy vocalizations will yield a single, highly accurate hit point anchored to the sharpest initial rise in spectral energy.

Recommended Parameter Ranges for Vocal Percussion

The ultimate efficacy of the "Beatbox2MIDI" engine is dictated by strict parameter calibration. Because human vocal cords and articulators cannot physically reproduce the micro-second precision of an electronic drum machine, the parameters must be slightly relaxed compared to algorithms tracking digital polyphonic instruments, yet tight enough to capture fast rhythmic subdivision without merging hits.⁴⁷

Based upon deep MIR research tailored specifically to vocal percussion and adaptive thresholding dynamics, the following table locks down the exact mathematical defaults and acceptable variance ranges that must be hardcoded or exposed within the C++ DSP engine:

Parameter	Mathematical Symbol	Recommended Default	Acceptable Range	Justification for Vocal Percussion
FFT Window Size	N	1024 samples ($\approx$ ms)	512 – 2048	Balances the sub-bass frequency resolution required for /b/ kicks with the temporal sharpness necessary to prevent the smearing of /ts/ hi-hats. ¹⁰
Hop Size	H	256 samples ($\approx$ ms)	128 – 512	Ensures 75% frame overlap with an

				$N = 1024$ window. At 44.1 kHz, a 256-sample hop provides sub-10ms timing resolution, critical for accurate downstream MIDI quantization. ¹⁵
Log Compression Factor	γ	10.0	1.0 – 100.0	Mathematically equalizes the vast amplitude disparity between thunderous vocal kicks and whispered, breathy hi-hats prior to the flux subtraction. ⁶
Adaptive Window Size	W	17 frames ($\approx$ ms)	9 – 35 frames	A 100-millisecond moving median window perfectly encapsulates the localized ambient noise of a single musical beat without accidentally encompassing adjacent 16th

				notes at tempos above 120 BPM. ³³
Sensitivity Multiplier	α	1.5	1.0 – 3.0	Scales the local median threshold dynamically. A default of 1.5 mandates that the energy delta must be 50% larger than the surrounding median noise floor to trigger a hit. ³⁶
Noise Floor Offset	δ	0.02	0.005 – 0.1	A flat mathematical floor added to the median threshold to completely block micro-fluctuations in empty digital space or background room tone from triggering false hits. ³³
Minimum Peak Distance	$T_{refractor}$	40 ms ($\approx$ frames)	25 ms – 80 ms	A 40ms lockout is brief enough to permit rapid 32nd-note drum rolls, but enduring enough to

				block the jagged, chaotic secondary peaks characteristic of a drawn-out vocal snare or lip oscillation. ⁴³
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Synthesis of the Comprehensive C++ DSP Pipeline

To synthesize the preceding mathematical analyses into an actionable engineering directive, the entire C++ architecture required to convert a floating-point array of raw beatboxing audio into a discrete array of hit-point sample indices is defined continuously as follows.

The engine must operate in sequential, rigidly separated phases to ensure maximum cache locality and deterministic execution within WebAssembly.

First, the initialization phase occurs. The engine receives the length of the audio buffer and pre-calculates the necessary dimensions for the state arrays. The Hann window coefficients array of size $N = 1024$ is populated utilizing the cosine formula. A prev_magnitude array of size 513 (accounting for the Nyquist limit of the real-to-complex FFT) is initialized strictly to 0.0 . An empty sf_envelope array is allocated to hold the resulting Novelty Function.

The core STFT loop initiates, advancing a read pointer by $H = 256$ samples until the termination of the audio array. Within this loop, N samples are extracted, multiplied element-wise by the Hann window, and processed via a dependency-free, lightweight Radix-2 Real-to-Complex FFT. The linear magnitude of the resulting complex bins is calculated, followed immediately by the application of the logarithmic compression function utilizing $\gamma = 10.0$. The L_1 Spectral Flux is computed by subtracting the prev_magnitude from the compressed magnitude, passing the result through the half-wave rectification maximum function, and summing the absolute differences across all k bins. This single scalar flux value is pushed to the sf_envelope array, and the prev_magnitude array is overwritten with the current frame's compressed magnitude to prepare for the subsequent hop.

Once the sf_envelope is fully populated, the threshold generation phase begins. The algorithm iterates over the sf_envelope utilizing a moving window of size $W = 17$. For each frame, it extracts the exact median of the local window using an efficient partial sorting algorithm. The

sensitivity multiplier $\alpha = 1.5$ and the noise floor offset $\delta = 0.02$ are applied to this median, and the resultant value is stored in a parallel threshold_envelope array.

Finally, the peak-picking and refractory phase executes. The algorithm iterates over the sf_envelope, comparing each frame against its corresponding value in the threshold_envelope. It validates the presence of a local maximum by checking the strictly greater-than status of the immediately adjacent frames. If the frame qualifies as a peak, the algorithm checks the frame distance against the state tracker last_onset_frame to ensure the $M_{refractory} = 7$ frames lockout period has expired. If all logical conditions evaluate to true, the onset is formally registered. The exact time-domain sample index is calculated using the mapping equation $i = m \cdot 256 + 512$, and this index is pushed to the final output array.

By decisively bypassing convolutional neural networks and deep learning mechanisms, and relying exclusively on this highly deterministic, acoustically optimized DSP architecture, the "Beatbox2MIDI" engine achieves robust, repeatable segmentation. The mathematical imperative to utilize the L1 norm explicitly addresses the broadband, low-amplitude nature of vocal fricatives, while the moving median adaptive threshold seamlessly tracks the extreme dynamic variance inherent to the human vocal tract. Coupled with strict sample-level refractory logic, this blueprint provides an exhaustively reliable mathematical foundation ready for immediate compilation into a high-performance WebAssembly module.

Works cited

1. Paralinguistic mechanisms of production in human "beatboxing": A real-time magnetic resonance imaging study - PMC, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3574116/>
2. Exploring the Spectral and Temporal Characteristics of Human Beatbox Sounds: A Preliminary Study | Request PDF - ResearchGate, accessed April 23, 2026, https://www.researchgate.net/publication/357249360_Exploring_the_Spectral_and_Temporal_Characteristics_of_Human_Beatbox_Sounds_A_Preliminary_Study
3. A Tutorial on Onset Detection in Music Signals, accessed April 23, 2026, https://www.iro.umontreal.ca/~pift6080/H09/documents/papers/bello_onset_tutorial.pdf
4. State-Of-The-Art for Audio Onset Detection // Week 3 | Music Information Retrieval, accessed April 23, 2026, <https://musicinformationretrieval.wordpress.com/2017/02/03/state-of-the-art-for-audio-onset-detection-week-3/>
5. Chapter 6: Tempo and Beat Tracking - AudioLabs Erlangen, accessed April 23, 2026, <https://www.audiolabs-erlangen.de/resources/MIR/FMP/C6/C6.html>
6. Beatboxing to Drums using Support Vector Machines - Hajim School of Engineering & Applied Sciences, accessed April 23, 2026, https://hajim.rochester.edu/ece/sites/zduan/teaching/ece477/projects/2017/Arvind_Ramanathan_ReportFinal.pdf

7. Wearable Haptic Music Player with Multi-Feature Extraction Using Spectral Flux and Yin Algorithms - MDPI, accessed April 23, 2026,
<https://www.mdpi.com/2079-9292/14/18/3658>
8. What does changing the window size of an FFT over time do to the frequency spectrum? : r/DSP - Reddit, accessed April 23, 2026,
https://www.reddit.com/r/DSP/comments/p4ld0a/what_does_changing_the_window_size_of_an_fft_over/
9. FFT window length, accessed April 23, 2026,
<https://www.speech.kth.se/snack/fftwin.html>
10. Introduction - Choosing the Right Window Size - Ircam, accessed April 23, 2026,
<https://support.ircam.fr/docs/AudioSculpt/3.0/co/The%20Right%20FFT.html>
11. Onset Detection Revisited - ResearchGate, accessed April 23, 2026,
https://www.researchgate.net/publication/200806116_Onset_Detection_Revisited
12. Superflux onsets — librosa-gallery 0.1.0 documentation, accessed April 23, 2026,
https://librosa.org/librosa_gallery/auto_examples/plot_superflux.html
13. DIFFERENTIABLE MODELLING OF PERCUSSIVE AUDIO WITH TRANSIENT AND SPECTRAL SYNTHESIS - Jordie Shier, accessed April 23, 2026,
https://jordieshier.com/assets/pdf/shier2023differentiable_paper.pdf
14. EVALUATING THE ONLINE CAPABILITIES OF ONSET DETECTION METHODS - ISMIR, accessed April 23, 2026,
<https://archives.ismir.net/ismir2012/paper/000049.pdf>
15. Beatbox Classification to Distinguish User Experiences Using Machine Learning Approaches - Science Publications, accessed April 23, 2026,
<https://thescipub.com/pdf/jcssp.2025.961.970.pdf>
16. audio - Whats the optimal window function to use for analyzing real-time data samples?, accessed April 23, 2026,
<https://dsp.stackexchange.com/questions/586/whats-the-optimal-window-function-to-use-for-analyzing-real-time-data-samples>
17. What's the ideal FFT window for measuring a group of signals of differing amplitudes but close in frequency?, accessed April 23, 2026,
<https://dsp.stackexchange.com/questions/71947/whats-the-ideal-fft-window-for-measuring-a-group-of-signals-of-differing-amplit>
18. OnsetDetection — Essentia 2.1-beta6-dev documentation, accessed April 23, 2026,
https://essentia.upf.edu/reference/streaming_OnsetDetection.html
19. Essentia - MTG projects, accessed April 23, 2026,
<https://mtg.github.io/essentia.js/docs/api/Essentia.html>
20. ONSET DETECTION REVISITED Simon Dixon Austrian Research Institute for Artificial Intelligence Freyung 6/6, Vienna 1010, Austria, accessed April 23, 2026,
<https://ofai.at/papers/oefai-tr-2006-12.pdf>
21. spectralFlux - Spectral flux for audio signals and auditory spectrograms - MATLAB, accessed April 23, 2026,
<https://la.mathworks.com/help/audio/ref/spectralflux.html>
22. Introduction Overview Onset Detection Feature Extraction Future Work Results SVM, accessed April 23, 2026,
<https://hajim.rochester.edu/ece/sites/zduan/teaching/ece477/projects/2017/Arvind>

[_Ramanathan_Poster.pdf](#)

23. Signal Processing Methods for Audio Classification and Music Content Analysis, accessed April 23, 2026, <https://cris.tuni.fi/ws/portalfiles/portal/2308383/eronen.pdf>
24. Why L1 norm creates Sparsity compared with L2 norm | by Satishkumar Moparthy - Medium, accessed April 23, 2026, <https://satishkumarmoparthy.medium.com/why-l1-norm-creates-sparsity-compared-with-l2-norm-3c6fa9c607f4>
25. L1-norm vs. L2-norm fitting in optimizing focal multi-channel tES stimulation: linear and semidefinite programming vs. weighted least squares - PubMed, accessed April 23, 2026, <https://pubmed.ncbi.nlm.nih.gov/36099674/>
26. Three Dimensions of Pitched Instrument Onset Detection, accessed April 23, 2026, https://www.csd.uoc.gr/~hannover/MMILab-Andre_files/HolzapfelIEEEOnset.pdf
27. [D] Why is L2 preferred over L1 Regularization? : r/MachineLearning - Reddit, accessed April 23, 2026, https://www.reddit.com/r/MachineLearning/comments/dgog2h/d_why_is_l2_preferred_over_l1_regularization/
28. Musical note onset detection based on a spectral sparsity measure - PMC - NIH, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8550344/>
29. MAXIMUM FILTER VIBRATO SUPPRESSION FOR ONSET DETECTION | Request PDF, accessed April 23, 2026, https://www.researchgate.net/publication/266556929_MAXIMUM_FILTER_VIBRATO_SUPPRESSION_FOR_ONSET_DETECTION
30. Maximum Filter Vibrato Suppression for Onset Detection - PHENICX, accessed April 23, 2026, https://phenicx.upf.edu/system/files/publications/Boeck_DAFx-13.pdf
31. Local Group Delay based Vibrato and Tremolo Suppression for Onset Detection - ISMIR, accessed April 23, 2026, <https://archives.ismir.net/ismir2013/paper/000175.pdf>
32. Analysing Multi-person Timing in Music and Movement: Event Based Methods in - Brill, accessed April 23, 2026, https://brill.com/display/book/edcoll/9789004280205/B9789004280205_009.xml
33. Full article: Musical robot swarms, timing, and equilibria - Taylor & Francis, accessed April 23, 2026, <https://www.tandfonline.com/doi/full/10.1080/09298215.2021.1910313>
34. ONLINE REAL-TIME ONSET DETECTION WITH RECURRENT NEURAL NETWORKS | Request PDF - ResearchGate, accessed April 23, 2026, https://www.researchgate.net/publication/266556788_ONLINE_REAL-TIME_ONSET_DETECTION_WITH_RECURRENT_NEURAL_NETWORKS
35. (PDF) Delayed Decision-making in Real-time Beatbox Percussion Classification, accessed April 23, 2026, https://www.researchgate.net/publication/252066013_Delayed_Decision-making_in_Real-time_Beatbox_Percussion_Classification
36. algorithm - Peak signal detection in realtime timeseries data - Stack Overflow,

accessed April 23, 2026,

<https://stackoverflow.com/questions/22583391/peak-signal-detection-in-realtime-timeseries-data>

37. Adaptive Threshold Generation for Fault Detection with High Dependability for Cyber-Physical Systems - MDPI, accessed April 23, 2026, <https://www.mdpi.com/2076-3417/8/11/2235>
38. PSYCHOACOUSTICS-WEB: A free online tool for the estimation of auditory thresholds, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11362506/>
39. An Asynchronous Delta Modulator for Spike Encoding in Event-Driven Brain-Machine Interface - arXiv, accessed April 23, 2026, <https://arxiv.org/html/2604.08758v2>
40. US7430254B1 - Matched detector/channelizer with adaptive threshold - Google Patents, accessed April 23, 2026, <https://patents.google.com/patent/US7430254B1/en>
41. Hormonal computing: a conceptual approach - PMC - NIH, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC10469008/>
42. Peak Picking - International Audio Laboratories Erlangen, accessed April 23, 2026, https://www.audiolabs-erlangen.de/resources/MIR/FMP/C6/C6S1_PeakPicking.html
43. Real-Time Detection of Industrial Respirator Fit Using Embedded Breath Sensors and Machine Learning Algorithms - PMC, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC12650192/>
44. Experimental Evaluation of Methods for Real-Time EEG Phase-Specific Transcranial Magnetic Stimulation - PMC, accessed April 23, 2026, <https://pmc.ncbi.nlm.nih.gov/articles/PMC8293904/>
45. A flexible quality metric for electrophysiological recordings across brain regions and species, accessed April 23, 2026, <https://www.biorxiv.org/content/10.64898/2026.03.06.710130v1.full-text>
46. Validating event-based vision sensor characterization methods through simulation - SPIE Digital Library, accessed April 23, 2026, <https://www.spiedigitallibrary.org/conference-proceedings-of-spie/13619/136191H/Validating-event-based-vision-sensor-characterization-methods-through-simulation/10.1117/12.3064703.full>
47. DESIGN AND EVALUATION OF ONSET DETECTORS USING DIFFERENT FUSION POLICIES - ISMIR, accessed April 23, 2026, <https://archives.ismir.net/ismir2014/paper/000229.pdf>
48. Towards Human-in-the-loop Onset Detection: A Transfer Learning Approach for Maracatu, accessed April 23, 2026, <https://arxiv.org/html/2507.04858>